

# RYAN PRATT

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## EDUCATION

### University of Wisconsin–Madison

B.S. in Computer Engineering; Additional Major in Computer Sciences | Expected May 2028 | GPA: 3.73/4.0

Relevant Coursework: Machine Organization & Programming, Digital System Fundamentals, Circuit Analysis, Signals, Information & Computation

In Progress: Operating Systems, Digital System Design & Synthesis, High Performance Computing

## SKILLS

**Languages:** C, C++, Python, Verilog, Java, Bash

**Tools/Platforms:** Linux, Git, MATLAB, Maven, JUnit, Mockito, SonarQube, Docker

**Libraries:** NumPy, SciPy, PyTorch

**Concepts:** Computer Architecture, Digital Logic, Algorithms, Unit Testing, Software Debugging, Hardware Simulation

## EXPERIENCE

### Wintrust Financial Corporation — Full Stack Engineer Intern

May 2026 – August 2026

- Increased backend code coverage from 16% to 41% across a roughly 76,000-line Java codebase, developing unit tests with JUnit, Mockito, and AEM Mocks for components, services, and servlets.
- Resolved 400+ SonarQube findings, including critical and major issues, by updating production Java code and deprecated APIs, incorporating peer review feedback, and documenting changes for QA.
- Developed a Java OSGi service for optimized image delivery, reducing image payload sizes by approximately 30% through WebP integration.

### iCode — Lead Instructor

June 2025 – August 2025

- Taught over 200 K–12 students programming and robotics through hands-on projects and guiding debugging.

## TECHNICAL EXPERIENCE

### Wisconsin Robotics — Software Team Member

October 2024 – December 2025

- Implemented A path planning in Python\* for rover navigation, reducing time-to-goal by approximately 15% in simulation and physical testing.
- Integrated navigation software for rover deployment, balancing algorithm performance with constrained hardware resources.

### Wisconsin AI Safety Initiative — Technical Safety Scholar

February 2025 – February 2026

- Analyzed approximately 60 AI research papers and model releases during a 13-week cohort, evaluating model capabilities, robustness, and failure modes.
- Compared alignment and interpretability approaches across academic and open-source systems, communicating technical tradeoffs and potential mitigation strategies in peer discussions.

## PROJECTS

### Photon-Sim — Photonic AI Accelerator Simulator

September 2025 – March 2026

**Tech:** Python (NumPy/SciPy), FastAPI, React (TypeScript), PyTorch, Tailwind CSS

- Built a photonic AI accelerator simulator using Python, NumPy, and SciPy, with a FastAPI backend and React/TypeScript dashboard for evaluating ML workloads against CPU baselines.
- Modeled signal loss, noise, and temperature drift, running MNIST inference to examine how simulated hardware conditions and numerical precision affect model accuracy.
- Automated benchmark sweeps across matrix sizes 64–1024, creating a CLI runner and reproducible CSV/JSON exports to compare accuracy, latency, and efficiency across simulated configurations.

### Amino Acid Substitution Prediction — Machine Learning Marathon 2025

September 2025 – December 2025

**Tech:** Python, PyTorch, XGBoost, pandas, NumPy, Jupyter Notebook

- Led a five-person team in a UW–Madison machine learning competition using 175K+ samples, analyzing amino acid substitution data to guide preprocessing and feature selection.
- Reduced cross-validation MSE by 23% by identifying heavy-tailed prediction errors and redesigning training to prioritize high-error cases.
- Compared model variants through cross-validation, selected a more stable final approach, and presented the modeling pipeline, validation strategy, and tradeoffs to 80+ attendees.